

Direct Products of Groups

Road map

This chapter begins with the First Isomorphism Theorem, which explains why a kernel naturally produces a quotient. We then construct the external direct product $G_1 \times G_2$, analyze its presentations and canonical factor subgroups, and prove the internal direct-product criterion. The final sections determine exactly when a product of finite cyclic groups is cyclic and record how subgroups, quotients, conjugacy classes, centralizers, and centers behave under products.

9.1 From a homomorphism to a quotient

Let

$$\varphi : G \longrightarrow G'$$

be a group homomorphism. Its kernel is a normal subgroup of G :

$$\ker \varphi \trianglelefteq G.$$

The quotient $G/\ker \varphi$ is therefore defined. Recall also that if $H \trianglelefteq G$, then

$$(aH)^{-1} = a^{-1}H,$$

because

$$(aH)(a^{-1}H) = aa^{-1}H = H,$$

the identity element of G/H .

The natural question is whether the quotient by the kernel recovers the part of G' actually reached by φ . The answer is yes.

Theorem 9.1: First Isomorphism Theorem

Let $\varphi : G \rightarrow G'$ be a group homomorphism. Define

$$\overline{\varphi} : G/\ker \varphi \longrightarrow \operatorname{im} \varphi, \quad \overline{\varphi}(a \ker \varphi) = \varphi(a).$$

Then $\bar{\varphi}$ is an isomorphism. Hence

$$G / \ker \varphi \cong \text{im } \varphi.$$

证明. We check every required point.

Homomorphism. For $a, b \in G$,

$$\begin{aligned} \bar{\varphi}((a \ker \varphi)(b \ker \varphi)) &= \bar{\varphi}(ab \ker \varphi) \\ &= \varphi(ab) \\ &= \varphi(a)\varphi(b) \\ &= \bar{\varphi}(a \ker \varphi)\bar{\varphi}(b \ker \varphi). \end{aligned}$$

Well-definedness and injectivity. For $a, b \in G$,

$$\begin{aligned} a \ker \varphi = b \ker \varphi &\iff a^{-1}b \in \ker \varphi \\ &\iff \varphi(a^{-1}b) = e_{G'} \\ &\iff \varphi(a)^{-1}\varphi(b) = e_{G'} \\ &\iff \varphi(a) = \varphi(b). \end{aligned}$$

The forward implication shows that the displayed definition does not depend on the representative of the coset. The reverse implication shows that two cosets with the same image are equal, so $\bar{\varphi}$ is injective.

Surjectivity onto the image. If $a' \in \text{im } \varphi$, then $a' = \varphi(a)$ for some $a \in G$, and

$$\bar{\varphi}(a \ker \varphi) = a'.$$

Thus $\bar{\varphi}$ is surjective onto $\text{im } \varphi$, completing the proof. ■

9.2 The external direct product

Definition 9.1: Direct product of two groups

Let G_1 and G_2 be groups. Their Cartesian product is

$$G_1 \times G_2 = \{(a_1, a_2) : a_1 \in G_1, a_2 \in G_2\}.$$

Define multiplication component by component:

$$\begin{aligned} \mu : (G_1 \times G_2) \times (G_1 \times G_2) &\longrightarrow G_1 \times G_2, \\ ((a_1, a_2), (b_1, b_2)) &\longmapsto (a_1 b_1, a_2 b_2). \end{aligned}$$

The resulting group is called the *external direct product* of G_1 and G_2 .

Proposition 9.1

The componentwise multiplication above makes $G_1 \times G_2$ a group. Its identity and inverse are

$$e_{G_1 \times G_2} = (e_1, e_2), \quad (a_1, a_2)^{-1} = (a_1^{-1}, a_2^{-1}).$$

证明. For three elements,

$$\begin{aligned} ((a_1, a_2)(b_1, b_2))(c_1, c_2) &= (a_1 b_1, a_2 b_2)(c_1, c_2) \\ &= ((a_1 b_1)c_1, (a_2 b_2)c_2) \\ &= (a_1(b_1 c_1), a_2(b_2 c_2)) \\ &= (a_1, a_2)((b_1, b_2)(c_1, c_2)), \end{aligned}$$

so associativity follows from associativity in each factor. Also,

$$(e_1, e_2)(a_1, a_2) = (a_1, a_2) = (a_1, a_2)(e_1, e_2),$$

and

$$(a_1, a_2)(a_1^{-1}, a_2^{-1}) = (e_1, e_2) = (a_1^{-1}, a_2^{-1})(a_1, a_2).$$

■

9.2.1 Immediate properties

1. If both groups are finite, then

$$|G_1 \times G_2| = |G_1| |G_2|.$$

This is the ordinary counting rule for a Cartesian product.

2. The product $G_1 \times G_2$ is Abelian if and only if both G_1 and G_2 are Abelian. Indeed,

$$(a_1, a_2)(b_1, b_2) = (b_1, b_2)(a_1, a_2)$$

for all pairs exactly when $a_1 b_1 = b_1 a_1$ and $a_2 b_2 = b_2 a_2$ for all components.

3. The sets $G_1 \times G_2$ and $G_2 \times G_1$ are generally not literally equal, but the swap map

$$(a_1, a_2) \mapsto (a_2, a_1)$$

is an isomorphism. Hence $G_1 \times G_2 \cong G_2 \times G_1$.

4. Direct products are associative up to the natural isomorphism

$$((a_1, a_2), a_3) \mapsto (a_1, (a_2, a_3)).$$

Thus one normally writes $G_1 \times G_2 \times G_3$ without parentheses.

Example 9.1: $\mathbb{Z} \times \mathbb{Z}$

In additive notation,

$$\mathbb{Z} \times \mathbb{Z} = \{(m_1, m_2) : m_1, m_2 \in \mathbb{Z}\},$$

with

$$(m_1, m_2) + (n_1, n_2) = (m_1 + n_1, m_2 + n_2).$$

This is the integer lattice in the plane under vector addition.

Example 9.2: $\mathbb{Z}_2 \times \mathbb{Z}_2$ and the Klein four group

Write the two copies of $\mathbb{Z}_2$ multiplicatively as

$$\mathbb{Z}_2^{(1)} = \{e_1, a_1\}, \quad \mathbb{Z}_2^{(2)} = \{e_2, a_2\},$$

with $a_1^2 = e_1$ and $a_2^2 = e_2$. The product has four elements:

$$(e_1, e_2), \quad (a_1, e_2), \quad (e_1, a_2), \quad (a_1, a_2).$$

Its Cayley table is

$\cdot$	(e_1, e_2)	(a_1, e_2)	(e_1, a_2)	(a_1, a_2)
(e_1, e_2)	(e_1, e_2)	(a_1, e_2)	(e_1, a_2)	(a_1, a_2)
(a_1, e_2)	(a_1, e_2)	(e_1, e_2)	(a_1, a_2)	(e_1, a_2)
(e_1, a_2)	(e_1, a_2)	(a_1, a_2)	(e_1, e_2)	(a_1, e_2)
(a_1, a_2)	(a_1, a_2)	(e_1, a_2)	(a_1, e_2)	(e_1, e_2)

Every nonidentity element has order 2. Under

$$(e_1, e_2) \mapsto e, \quad (a_1, e_2) \mapsto a, \quad (e_1, a_2) \mapsto b, \quad (a_1, a_2) \mapsto c,$$

this table becomes

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

the Cayley table of the Klein four group $K_4 = V_4$. Therefore

$$\boxed{\mathbb{Z}_2 \times \mathbb{Z}_2 \cong K_4.}$$

Example 9.3: $\mathbb{Z}_3 \times \mathbb{Z}_2$

Let

$$G_1 = \langle a : a^3 = e_1 \rangle \cong \mathbb{Z}_3, \quad G_2 = \langle b : b^2 = e_2 \rangle \cong \mathbb{Z}_2.$$

Using the abbreviations

$$a = (a, e_2), \quad b = (e_1, b),$$

the product is

$$G_1 \times G_2 = \{e, a, a^2, b, ab, a^2b\} = \langle a, b : a^3 = e, b^2 = e, ab = ba \rangle.$$

The relation $ab = ba$ is necessary because the two canonical factors commute. The element

a^2b generates the entire product:

$$\begin{array}{lll} (a^2b)^1 = a^2b, & (a^2b)^2 = a, & (a^2b)^3 = b, \\ (a^2b)^4 = a^2, & (a^2b)^5 = ab, & (a^2b)^6 = e. \end{array}$$

Hence

$$\mathbb{Z}_3 \times \mathbb{Z}_2 \cong \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3.$$

9.3 Presentations of direct products

Suppose

$$G_1 = \langle S_1 : R_1 \rangle, \quad G_2 = \langle S_2 : R_2 \rangle,$$

where the generating sets S_1 and S_2 are disjoint. In the direct product, every generator from the first factor commutes with every generator from the second. Thus

$$G_1 \times G_2 = \langle S_1 \cup S_2 : R_1 \cup R_2 \cup \{st = ts : s \in S_1, t \in S_2\} \rangle.$$

The original relations R_1 and R_2 control each factor, while the additional cross-relations force the two factors to commute.

9.4 The canonical factor subgroups

Let e_1, e_2 be the identity elements of G_1, G_2 . Define

$$G'_1 = G_1 \times \{e_2\} = \{(a_1, e_2) : a_1 \in G_1\},$$

$$G'_2 = \{e_1\} \times G_2 = \{(e_1, a_2) : a_2 \in G_2\}.$$

The maps

$$G_1 \longrightarrow G'_1, \quad a_1 \longmapsto (a_1, e_2),$$

and

$$G_2 \longrightarrow G'_2, \quad a_2 \longmapsto (e_1, a_2),$$

are isomorphisms. Hence $G'_1 \cong G_1$ and $G'_2 \cong G_2$, and both are subgroups of $G_1 \times G_2$.

Theorem 9.2: Canonical factors are normal

The canonical factor subgroups satisfy

$$G'_1 \trianglelefteq G_1 \times G_2, \quad G'_2 \trianglelefteq G_1 \times G_2,$$

and

$$(G_1 \times G_2)/G'_1 \cong G_2, \quad (G_1 \times G_2)/G'_2 \cong G_1.$$

证明. Consider the projection

$$\pi_2 : G_1 \times G_2 \longrightarrow G_2, \quad \pi_2(a_1, a_2) = a_2.$$

It is a homomorphism because

$$\pi_2((a_1, a_2)(b_1, b_2)) = \pi_2(a_1b_1, a_2b_2) = a_2b_2 = \pi_2(a_1, a_2)\pi_2(b_1, b_2).$$

Its kernel is

$$\ker \pi_2 = \{(a_1, e_2) : a_1 \in G_1\} = G'_1,$$

so $G'_1 \trianglelefteq G_1 \times G_2$. Since π_2 is surjective, the First Isomorphism Theorem gives

$$(G_1 \times G_2)/G'_1 \cong G_2.$$

Interchanging the roles of the two factors proves the other two statements. ■

It is instructive to see the quotient cosets directly. For $(a_1, a_2) \in G_1 \times G_2$,

$$\begin{aligned} (a_1, a_2)G'_1 &= \{(a_1, a_2)(g_1, e_2) : g_1 \in G_1\} \\ &= \{(a_1g_1, a_2) : g_1 \in G_1\} \\ &= G_1 \times \{a_2\}. \end{aligned}$$

This coset depends only on a_2 , not on a_1 . The map

$$(a_1, a_2)G'_1 \mapsto a_2$$

is therefore well defined. It is injective because equality of the two cosets is equivalent to equality of their second components, it is surjective because every $a_2 \in G_2$ occurs, and it is a homomorphism because coset multiplication multiplies the second components.

9.5 Recognizing an internal direct product

Inside $G_1 \times G_2$, the canonical copies have four conspicuous properties:

1. $G'_1 \cap G'_2 = \{(e_1, e_2)\}$;
2. $G'_1 G'_2 = G_1 \times G_2$;
3. every element of G'_1 commutes with every element of G'_2 ;
4. G'_1 and G'_2 are normal in $G_1 \times G_2$.

Indeed,

$$(a_1, e_2)(e_1, a_2) = (a_1, a_2) = (e_1, a_2)(a_1, e_2).$$

The next theorem shows that these properties characterize direct-product decompositions inside an arbitrary group.

Theorem 9.3: Internal direct-product criterion

Let $H, K \leq G$. Suppose

$$H \cap K = \{e\}, \quad HK = G, \quad H \trianglelefteq G, \quad K \trianglelefteq G.$$

Then every $h \in H$ commutes with every $k \in K$, and

$$\kappa : H \times K \longrightarrow G, \quad \kappa(h, k) = hk,$$

is an isomorphism. Thus

$$G \cong H \times K.$$

证明. We first prove the elementwise commutation that is needed for κ to be a homomorphism. Let $h \in H$ and $k \in K$. Normality of H gives

$$kh^{-1}k^{-1} \in H,$$

and hence

$$(hk)(kh)^{-1} = hkh^{-1}k^{-1} \in H.$$

Normality of K similarly gives

$$hkh^{-1} \in K,$$

so

$$(hk)(kh)^{-1} = hkh^{-1}k^{-1} \in K.$$

Therefore

$$(hk)(kh)^{-1} \in H \cap K = \{e\}.$$

It follows that $hk = kh$.

Now take $(h_1, k_1), (h_2, k_2) \in H \times K$. Using the commutation just proved,

$$\begin{aligned} \kappa((h_1, k_1)(h_2, k_2)) &= \kappa(h_1h_2, k_1k_2) \\ &= h_1h_2k_1k_2 \\ &= h_1k_1h_2k_2 \\ &= \kappa(h_1, k_1)\kappa(h_2, k_2). \end{aligned}$$

Thus κ is a homomorphism.

The equality $HK = G$ says that every $g \in G$ has the form $g = hk$, so κ is surjective. Finally, if $(h, k) \in \ker \kappa$, then $hk = e$, and therefore

$$k = h^{-1} \in H.$$

Since also $k \in K$, we have $k \in H \cap K = \{e\}$, so $k = e$ and $h = e$. Hence

$$\ker \kappa = \{(e, e)\},$$

and κ is injective. ■

Proposition 9.2: Equivalent commuting version

In the internal direct-product criterion, the two normality assumptions may be replaced by

$$hk = kh \quad (h \in H, k \in K).$$

Thus either of the following sets of assumptions is sufficient:

$$H \cap K = \{e\}, \quad HK = G, \quad H, K \trianglelefteq G,$$

or

$$H \cap K = \{e\}, \quad HK = G, \quad [H, K] = \{e\}.$$

证明. Only the normality implication needs proof. Assume $HK = G$ and that elements of H commute with elements of K . Write an arbitrary $g \in G$ as $g = h_0 k_0$. Then

$$\begin{aligned} gHg^{-1} &= h_0 k_0 H k_0^{-1} h_0^{-1} \\ &= h_0 H h_0^{-1} \\ &= H, \end{aligned}$$

because k_0 commutes with every element of H , and a subgroup is invariant under conjugation by its own elements. Hence $H \trianglelefteq G$. The same argument gives $K \trianglelefteq G$. ■

Example 9.4: An internal decomposition of K_4

Using the table above, let

$$H = \{e, a\} = \langle a \rangle, \quad K = \{e, b\} = \langle b \rangle.$$

The Klein group is Abelian, so $H, K \trianglelefteq K_4$. Moreover,

$$H \cap K = \{e\}, \quad HK = \{e, a, b, ab\} = \{e, a, b, c\} = K_4.$$

The internal direct-product criterion gives

$$K_4 \cong H \times K \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

9.6 Products of finite cyclic groups

Let

$$\mathbb{Z}_{m_1} = \langle g_1 \rangle, \quad \mathbb{Z}_{m_2} = \langle g_2 \rangle,$$

where $g_1^{m_1} = e_1$ and $g_2^{m_2} = e_2$. Then

$$|\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2}| = m_1 m_2.$$

For any positive integer k ,

$$(g_1, g_2)^k = (e_1, e_2)$$

if and only if $m_1 \mid k$ and $m_2 \mid k$. Therefore

$$\text{ord}(g_1, g_2) = \text{lcm}(m_1, m_2). \quad (9.1)$$

More generally,

$$\text{ord}(x_1, x_2) = \text{lcm}(\text{ord } x_1, \text{ord } x_2).$$

Theorem 9.4: When a product of two cyclic groups is cyclic

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1.$$

证明. If $\gcd(m, n) = 1$, then

$$\text{lcm}(m, n) = mn.$$

By (9.1), the element (g_1, g_2) has order mn , which equals the order of the entire product. It therefore generates the product, and the product is cyclic of order mn .

Conversely, if the product is cyclic, it contains an element of order mn . Every element (x_1, x_2) has order

$$\text{lcm}(\text{ord } x_1, \text{ord } x_2) \leq \text{lcm}(m, n).$$

Thus $mn \leq \text{lcm}(m, n)$. Since

$$\text{lcm}(m, n) = \frac{mn}{\gcd(m, n)} \leq mn,$$

equality must hold and $\gcd(m, n) = 1$. ■

In additive quotient notation, the explicit isomorphism in the coprime case is

$$\Phi : \mathbb{Z}/(mn)\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \quad [r]_{mn} \mapsto ([r]_m, [r]_n).$$

Its kernel consists of integers divisible by both m and n . When $\gcd(m, n) = 1$, this means divisibility by mn , so the induced map is injective; the finite groups on the two sides have the same order, so it is also surjective. This is the group-theoretic form of the Chinese Remainder Theorem.

Proposition 9.3: Several cyclic factors

If $m_1, \dots, m_r$ are pairwise coprime, then

$$\mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_r} \cong \mathbb{Z}_{m_1 \cdots m_r}.$$

证明. If g_i generates $\mathbb{Z}_{m_i}$, then

$$\text{ord}(g_1, \dots, g_r) = \text{lcm}(m_1, \dots, m_r).$$

Pairwise coprimality makes this least common multiple equal to the product $m_1 \cdots m_r$, which is also the order of the whole direct product. ■

Example 9.5

Because 3 and 4 are coprime,

$$\mathbb{Z}_{12} \cong \mathbb{Z}_3 \times \mathbb{Z}_4.$$

On the other hand,

$$\mathbb{Z}_4 \not\cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The group $\mathbb{Z}_4$ contains an element of order 4, whereas every nonidentity element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ has order 2.

9.7 Further properties of direct products

9.7.1 Subgroups, normal subgroups, and quotients

If $H_1 \leq G_1$ and $H_2 \leq G_2$, then

$$H_1 \times H_2 \leq G_1 \times G_2.$$

Closure and inverses are checked componentwise. If, in addition,

$$H_1 \trianglelefteq G_1, \quad H_2 \trianglelefteq G_2,$$

then

$$H_1 \times H_2 \trianglelefteq G_1 \times G_2,$$

because

$$\begin{aligned} (g_1, g_2)(h_1, h_2)(g_1, g_2)^{-1} \\ = (g_1 h_1 g_1^{-1}, g_2 h_2 g_2^{-1}) \in H_1 \times H_2. \end{aligned}$$

Proposition 9.4: A quotient of a product

If $H_i \trianglelefteq G_i$ for $i = 1, 2$, then

$$\frac{G_1 \times G_2}{H_1 \times H_2} \cong \frac{G_1}{H_1} \times \frac{G_2}{H_2}.$$

证明. Define

$$\Psi : G_1 \times G_2 \longrightarrow G_1/H_1 \times G_2/H_2, \quad \Psi(g_1, g_2) = (g_1 H_1, g_2 H_2).$$

It is a surjective homomorphism, and

$$\begin{aligned} \ker \Psi &= \{(g_1, g_2) : g_1 \in H_1, g_2 \in H_2\} \\ &= H_1 \times H_2. \end{aligned}$$

The result follows from the First Isomorphism Theorem. ■

Not every subgroup is a product subgroup

For any group G , the diagonal

$$\Delta_G = \{(a, a) : a \in G\}$$

is a subgroup of $G \times G$, because

$$(a, a)(b, b) = (ab, ab), \quad (a, a)^{-1} = (a^{-1}, a^{-1}).$$

Except in trivial cases, it is not of the form $H \times H$: a product subgroup permits independent choices in the two coordinates, whereas the diagonal requires them to be equal.

9.7.2 Conjugacy classes and centralizers

Conjugation in a direct product is componentwise:

$$\begin{aligned} (g_1, g_2)(a_1, a_2)(g_1, g_2)^{-1} \\ = (g_1 a_1 g_1^{-1}, g_2 a_2 g_2^{-1}). \end{aligned}$$

It follows that each conjugacy class is a Cartesian product:

$$\text{Cl}_{G_1 \times G_2}(a_1, a_2) = \text{Cl}_{G_1}(a_1) \times \text{Cl}_{G_2}(a_2).$$

Similarly, (g_1, g_2) commutes with (a_1, a_2) exactly when g_1 commutes with a_1 and g_2 commutes with a_2 . Hence

$$\text{Cen}_{G_1 \times G_2}(a_1, a_2) = \text{Cen}_{G_1}(a_1) \times \text{Cen}_{G_2}(a_2).$$

Intersecting the centralizers of all elements gives the center:

$$\boxed{Z(G_1 \times G_2) = Z(G_1) \times Z(G_2).}$$

9.7.3 Products of more than two groups

For groups $G_1, \dots, G_r$, define

$$G_1 \times \cdots \times G_r = \{(a_1, \dots, a_r) : a_i \in G_i \text{ for } i = 1, \dots, r\},$$

with componentwise multiplication

$$(a_1, \dots, a_r)(b_1, \dots, b_r) = (a_1 b_1, \dots, a_r b_r).$$

Its identity and inverses are also componentwise:

$$e = (e_1, \dots, e_r), \quad (a_1, \dots, a_r)^{-1} = (a_1^{-1}, \dots, a_r^{-1}).$$

What to retain from this chapter

1. The First Isomorphism Theorem states $G/\ker \varphi \cong \text{im } \varphi$.
2. The direct product $G_1 \times G_2$ uses componentwise multiplication, identity, and inverses; its order is $|G_1||G_2|$ when the factors are finite.
3. The canonical factor copies are normal, and quotienting by one recovers the other factor.
4. If $H, K \trianglelefteq G$, $H \cap K = \{e\}$, and $HK = G$, then the factors commute and $G \cong H \times K$. Elementwise commutation can replace the two normality assumptions.
5. The product $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic of order mn exactly when $\gcd(m, n) = 1$.
6. Product subgroups and quotient groups behave componentwise, but subgroups such as the diagonal need not themselves be products.
7. Conjugacy classes, centralizers, and centers of direct products are the Cartesian products of the corresponding structures in the factors.